

NSLS2EM for Beamlines

Technical Manual

DSSI NSLS2EM Development Group

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Rev 0.2 draft version prepared by K. Ha (kha@bnl.gov), August 15, 2025



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1. Scope and Purpose

This manual provides a comprehensive technical description of the NSLS2EM system, covering both hardware architecture and software implementation. It is intended to serve as a reference for users, developers, and integrators working with NSLS2EM in beamline applications.

2. Hardware Overview

The **NSLS-II new electrometer** represents a next-generation, high-precision current measurement and feedback control platform developed to meet the demanding requirements of advanced X-ray beamline diagnostics. At the core of this system is the Avnet PicoZed 7030 System-on-Module (SoM), which integrates a Xilinx Zynq-7030 device—featuring dual-core ARM Cortex-A9 processors tightly coupled with programmable FPGA logic. This architecture provides a powerful and flexible framework for implementing real-time signal processing, control algorithms, and system-level coordination.

The electrometer is designed to support four high-resolution current input channels, each capable of measuring ultra-low to high current ranges. A precision analog front-end (ADA4350) followed by a 20-bit ADC (AD2378) enables accurate current measurement across six gain settings, covering input ranges from ± 100 nA to ± 10 mA, with effective resolution as low as 200 fA. This wide dynamic range ensures compatibility with both low-leakage and high-intensity detector outputs, making the system ideal for insertion device beamlines, diagnostics, and feedback applications at NSLS-II.

To support extended functionality, the system includes a daughterboard interface connector, allowing for modular expansion. A currently available daughterboard provides four voltage input channels, helpful in monitoring additional analog signals such as bias voltages or photodiode outputs. This modularity ensures the electrometer remains adaptable to future experimental demands.

A bipolar bias DAC (AD5060) is included to provide detector biasing over a wide voltage range of ± 12 V, supporting high-output current for driving sensors such as diamond detectors. In addition, a feedback DAC (AD5754) offers four analog outputs with a full-scale range of ± 10.8 V, designed to drive corrector magnets, piezo actuators, or other analog interfaces in real-time feedback loops.

2.1 Hardware Functional Descriptions

Current Input Channels ($\times 4$): Each channel uses an ADA4350 transimpedance amplifier followed by an AD2378 20-bit ADC. This chain provides precision measurement of input currents across six software-selectable gain settings. The channels support continuous simultaneous sampling at high resolution, enabling real-time beam current monitoring and feedback. The DAC output is provides selectable modes and default range is ± 10 V.

Bias DAC – AD5060: A ± 12 V-output, 16-bit DAC used to bias detectors or analog front-

ends. It supports high-output current drive, making it suitable for biasing low-leakage detectors such as diamond-based sensors. The bias output is programmable by PV, +/-10V range.

Feedback DAC – AD5754 ($\times 4$ outputs): Offers four bipolar analog outputs with a voltage range of ± 10.8 V. Typically used for closed-loop feedback, such as actuating corrector magnets or driving piezo actuators for position control.

General-Purpose I/O: A 16-pin header exposes 4 TTL-compatible digital inputs and 4 TTL outputs, which can be used for external triggers, status signaling, or interlock control.

Daughterboard Interface Connector: Supports custom or standard add-on boards for analog or digital expansion. Currently, a voltage input daughterboard is available for extending system capability to voltage signal acquisition.

PicoZed 7030 SoM – FPGA Core [3]: Executes all data acquisition, signal processing, and control logic. The Zynq FPGA combines hardware flexibility with ARM processor software programmability, making it suitable for real-time feedback and embedded networking.

SFP0/SFP1 Optical Transceivers: Provide high-speed optical communication, compatible with timing systems such as MRF Event System or long-distance streaming to supervisory controls.

Gigabit Ethernet / USB 2.0: Used for host communication, system updates, and remote diagnostics. Enables integration with EPICS control systems or custom TCP/UDP protocols.

microUSB Console: Provides UART-based terminal access to the embedded Linux OS for debugging, configuration, and logging.

SD Card Interface: Supports removable storage for OS boot, configuration files, and long-term data logging.

Low Voltage Regulators and Power Input: Internal regulators generate necessary supply rails from a single external +12 V input. Power sequencing is handled on board to ensure stable operation.

Figures 1 and 2 show the front and rear panel configurations of the hardware.

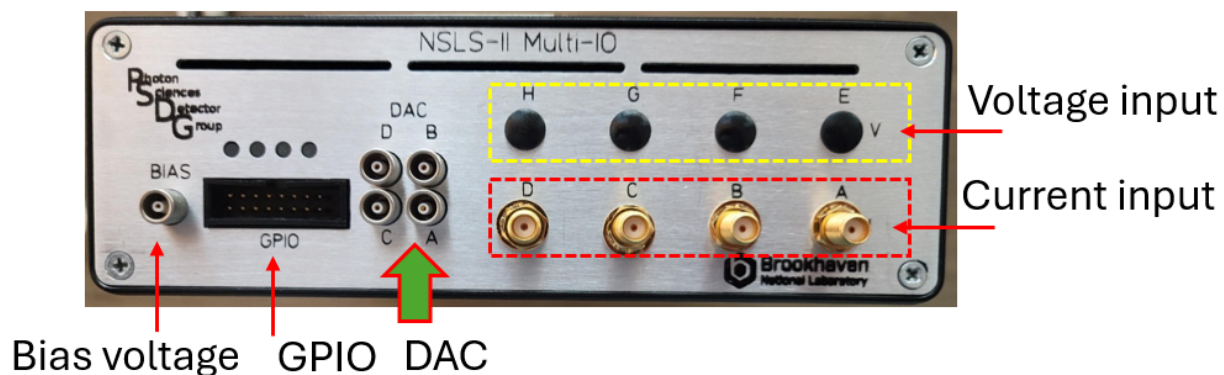


Figure 1: Front panel. from left: Bias voltage output, GPIO connector, 4 channel DAC, SAM for analog input

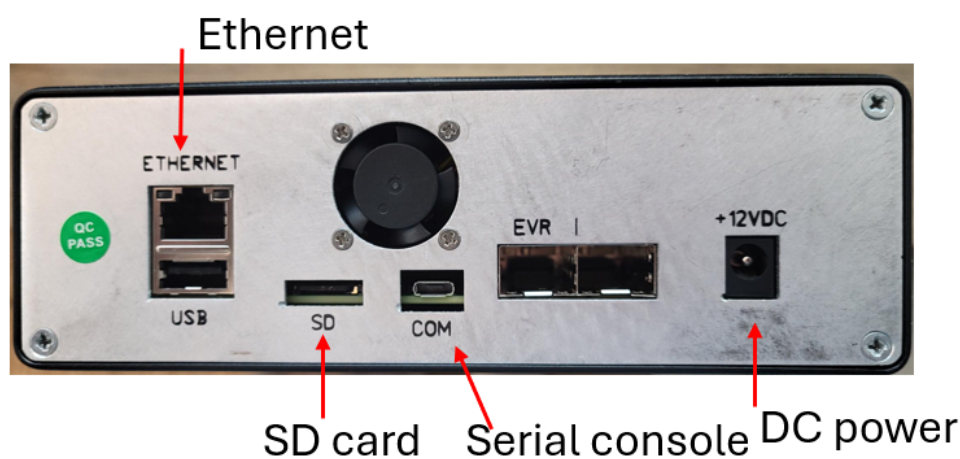


Figure 2: Rear panel.

2.2 Current Inputs

Verify that the output current range of the detector falls within the acceptable input current range of the NSLS2EM.

- 4-channel current input (SMA female-type connector), ADC max $F_s=378$ kHz
- Individual range: 100 nA–10 mA current input range selectable
- Adjustable Fast Acquisition (FA) update rate: 100 Hz to 10 kHz

Readout Channels	4 Current (independent gain control) 4 Voltage (independent gain control, via optional daughterboard)														
Current Measuring Range	<table> <tr> <th>Range</th><th>Resolution</th></tr> <tr> <td>± 100 nA</td><td>200 fA/bit</td></tr> <tr> <td>± 1 μA</td><td>2 pA/bit</td></tr> <tr> <td>± 10 μA</td><td>20 pA/bit</td></tr> <tr> <td>± 100 μA</td><td>200 pA/bit</td></tr> <tr> <td>± 1 mA</td><td>2 nA/bit</td></tr> <tr> <td>± 10 mA</td><td>20 nA/bit</td></tr> </table>	Range	Resolution	± 100 nA	200 fA/bit	± 1 μ A	2 pA/bit	± 10 μ A	20 pA/bit	± 100 μ A	200 pA/bit	± 1 mA	2 nA/bit	± 10 mA	20 nA/bit
Range	Resolution														
± 100 nA	200 fA/bit														
± 1 μ A	2 pA/bit														
± 10 μ A	20 pA/bit														
± 100 μ A	200 pA/bit														
± 1 mA	2 nA/bit														
± 10 mA	20 nA/bit														
ADC Sample Rate	Programmable, up to 1 MHz. It can be synchronized to the NSLS-II turn-by-turn event via the embedded event receiver. ~ 378 kHz														
Resolution Bits	20														
Bias Voltage Output	-10 V to +10 V														
Current Polarity	Bipolar														

Table 1: Analog signal specifications

2.3 General Purpose Input and Output (GPIO)

Four digital inputs and four digital outputs are provided via a 16-pin, 0.1-inch header to facilitate interfacing and triggering with other equipment on the beamline. The outputs are standard +5 V TTL levels. These I/O lines are mapped to the processor, allowing software to read from the inputs and control the outputs. Additional custom functionality can be implemented on these I/O pins upon request.

- 4 Digital TTL inputs (0 to 3.7V)
- 4 Digital TTL outputs (0 to 3.7V)
- Output individual adjustable frequency and duty cycle
- Output individual on/off control
- Set maximum count number to provide configured output clock and automatically stop after reaching the maximum clock number
- 4-channel clock output edge counts
- 4-channel status LED for:
 - Control mode (PV control/ISS beamline mode)
 - On/off status

Pin #	Function
1	Output 3
3	Output 2
5	Output 1
7	Output 0
9	Input 3
11	Input 2
13	Input 1
15	Input 0
2, 4, 6, 8, 10, 12, 14	Ground

Table 2: GPIO Connector Pin Assignment (top right is pin 1)

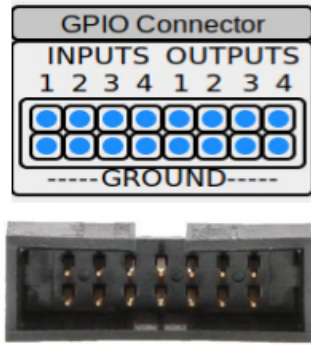


Figure 3: Current input control and monitoring page.

2.4 Digital to Analog Output (DAC)

Four DAC output channels are provided for precision motion feedback control, specifically designed to drive nanopositioning stages. Each DAC features 16-bit resolution and supports six software-programmable output voltage ranges, allowing for flexible adaptation to various stage requirements. The DAC offers a maximum settling time of 10 μ s and supports update rates of up to 378 kHz, ensuring fast and accurate signal delivery for high-performance closed-loop control.

2.5 Bias output

A detector bias is provided from an Analog Devices AD5060 16-bit DAC, which is followed by an OPA552 power amplifier, which can provide up to 200 mA of output current and ± 10 V ranges.

2.6 SFP

The Zynq-7030 includes high-speed GTX transceivers, which are used in NSLS2EM for serial communication interfaces such as multi-gigabit optical links. These transceivers support data rates up to 6.6 Gbps and provide reliable, low-latency transmission required for fast beamline feedback and diagnostics. The SFP0 port is used for the EVR timing interface, and SFP1 is used for the accelerator system data interface for the X-ray beam feedback system.

3. FPGA firmware

3.1 Current input signal processing

The Analog-to-Digital Converter (ADC) samples the input signal at a rate of 378 kHz. The resulting high-rate raw data is processed through a programmable averaging stage, which reduces noise and bandwidth to suit subsequent processing needs. The averaged result, hereafter referred to as **FA** (Filtered/Averaged data), serves two key roles within the system architecture.

First, the FA data is transmitted to the **quadEM** system for high-precision current and position monitoring. To support this data stream, FA samples are continuously written into a FIFO buffer, from which the embedded ARM processor retrieves the data in real time. The processor then forwards the data to the quadEM EPICS Input/Output Controller (IOC) via a software communication interface.

Second, the FA signal is simultaneously used as the input to a local Proportional-Integral (PI) feedback controller. This controller operates in real time to regulate beam stability based on the processed signal.

This dual-path utilization of the FA signal enables efficient and synchronized integration of both diagnostic and feedback functions within the electrometer system.

Let the original ADC data sampled at 378 kHz be represented as:

$$x[0], x[1], x[2], \dots, x[M - 1]$$

We define block averaging with a block size of N , where:

$$1 \leq N \leq 3800$$

The block-averaged output $y[k]$ is computed using the following equation:

$$y[k] = \frac{1}{N} \sum_{i=0}^{N-1} x[k \cdot N + i] \quad \text{for } k = 0, 1, 2, \dots, \left\lfloor \frac{M}{N} \right\rfloor - 1$$

Example: $N = 378$

For $N = 378$, the averaging reduces the output data rate to approximately 1 kHz:

$$y[k] = \frac{1}{378} \sum_{i=0}^{377} x[378k + i]$$

Remarks

- This block averaging is a form of finite impulse response (FIR) filtering with a rectangular window.
- It differs from a moving average in that it uses non-overlapping blocks.
- In FPGA hardware implementations, the sum is accumulated and then divided using either integer division by PV N .

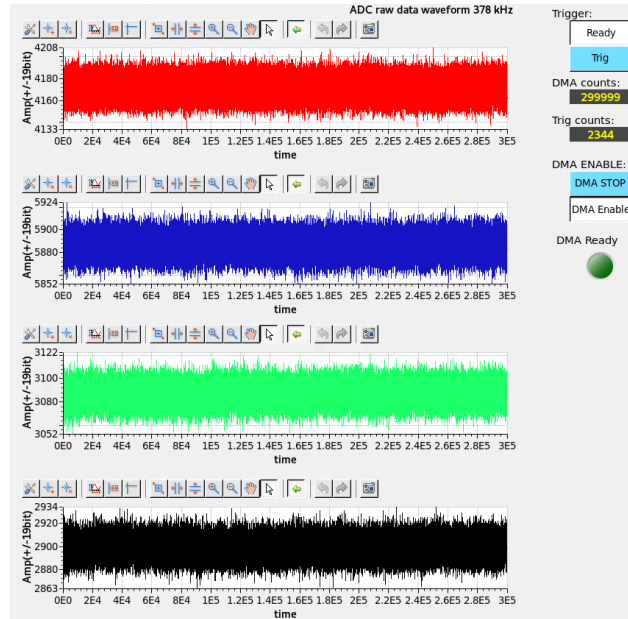


Figure 4: ADC current input waveform 378 kHz, 300 k points

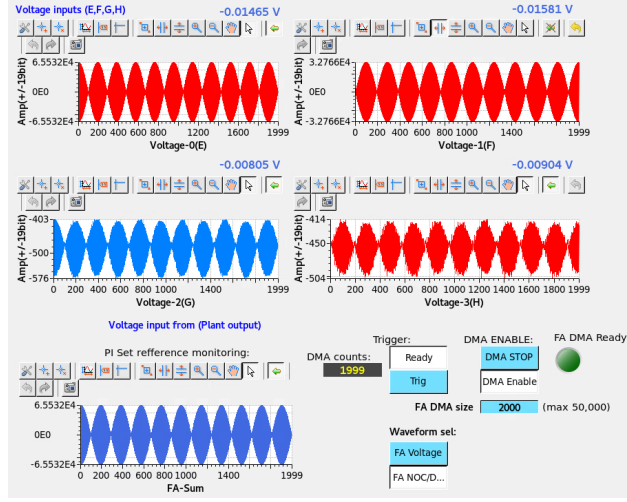


Figure 5: FA data waveform 10 kHz, 50 k points

3.2 X-ray beam position calculation

The FPGA firmware provides two different beam position calculation geometries, and users can select the appropriate geometry for each beamline. These geometries typically correspond to different electrode configurations, such as square (quad) or diagonal (diamond) modes, allowing optimal performance based on the beamline's specific detector setup. As a reference, the HXN (3-ID) beamline employs the square geometry mode in Hutch-B and the diamond geometry mode in Hutch-C, depending on the detector configuration.

1) QuadEM Diamond Geometry Calculation:

$$X = K_x \left(\frac{B - A}{A + B} \right) - X_{\text{offset}}$$

$$Y = K_y \left(\frac{D - C}{C + D} \right) - Y_{\text{offset}}$$

Where:

A = Channel 1

B = Channel 2

C = Channel 3

D = Channel 4

K_x, K_y = Calibration factors

$X_{\text{offset}}, Y_{\text{offset}}$ = Position offsets

2) QuadEM Square Mode position calculation:

$$X = K_x \left(\frac{(B + C) - (A + D)}{A + B + C + D} \right) - X_{\text{offset}}$$
$$Y = K_y \left(\frac{(A + B) - (C + D)}{A + B + C + D} \right) - Y_{\text{offset}}$$

Where:

A = Channel 1

B = Channel 2

C = Channel 3

D = Channel 4

K_x, K_y = Calibration factors

$X_{\text{offset}}, Y_{\text{offset}}$ = Position offsets

Sum = $A + B + C + D$

3.3 Event receiver for time synchronization

The EVR is widely used to synchronize the accelerator timing system and the RF system. At NSLS-II, the EVR is used to decode RF (499.654 MHz) synchronized Frev (378 kHz), Fast Acquisition (FA, 10 kHz), and Slow Acquisition (SA, 10 Hz) clocks for logic trigger inputs. The EVR provides timestamps received from the EVG. The EVR packet size is 16 bits, containing an 8-bit distribution bus and an 8-bit event code transmitted every 125 MHz clock cycle. The NSLS-II EVG configuration uses only the 8-bit event code field for timing event distribution. For the NSLS2EM ADC, the clock is selectable to an external clock from DBUS.0 Frev frequency, which is 378 kHz. Timestamp decoded by EVG and displays Unix timestamp; this hardware timestamp is used for synchronized data acquisition for the fly-scan applications.

If the EVR fiber optics cable is not available from accelerator timing system, it need to select internal clock mode only.

3.4 High speed data output

The high-speed data communication port uses a GTX port and is mainly designed for beam-line and accelerator system interface. For the HXN, this port is the Cell 3 accelerator fast orbit feedback interface.

Table 3: GTX Tx Packet Structure

Field	Bit Width	Description
Packet Length	12 bits [31:20]	Length of the packet in 32-bit words
Address	12 bits [[19:8]]	Device or channel address
Start Code	8 bits	Start marker (fixed value: 0x5C)
Ch-A	32 bits	Channel A measurement value
Ch-B	32 bits	Channel B measurement value
Ch-C	32 bits	Channel C measurement value
Ch-D	32 bits	Channel D measurement value
Pos-X	32 bits	Calculated X position from 4 channels
Pos-Y	32 bits	Calculated Y position from 4 channels
Sum	32 bits	Total signal (A + B + C + D)
Status	32 bits	Status or flag register
CRC	32 bits	CRC32 checksum for error checking

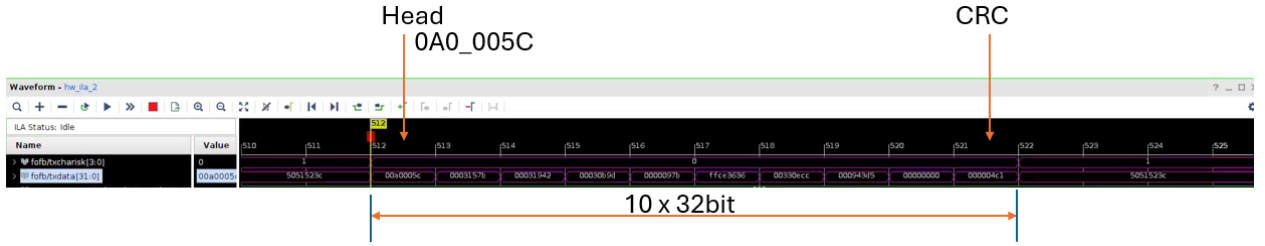


Figure 6: GTX Tx timing (FA data stream).

3.5 High speed PI feedback controller

The RTL PI function is testing at the HXN(3-ID) hutch-B for the local X-ray feedback implementation.

The NSLS2EM provides an FPGA RTL-based PI controller for fast local X-ray position feedback applications. This document describes the discrete-time PI (Proportional-Integral) controller implemented in an FPGA system. The sampling frequency is assumed to be:

$$F_s = 10 \text{ kHz}, \quad T_s = \frac{1}{F_s} = 100 \mu s$$

The FPGA operates at a clock frequency of:

$$F_{\text{clk}} = 10 \text{ MHz}$$

PI Controller Structure

Let:

- $E(n)$: Error signal at sample n
- $E(n-1)$: Previous error signal
- G_p : Proportional gain
- G_i : Integral gain
- $Y_d(n)$: Output of the integrator at sample n
- $Y(n)$: Final PI controller output at sample n

The PI controller output is defined as:

$$Y(n) = G_p \cdot E(n) + Y_d(n)$$

The integrator term uses the trapezoidal (Tustin) method:

$$Y_d(n) = Y_d(n-1) + G_i \cdot (E(n) + E(n-1))$$

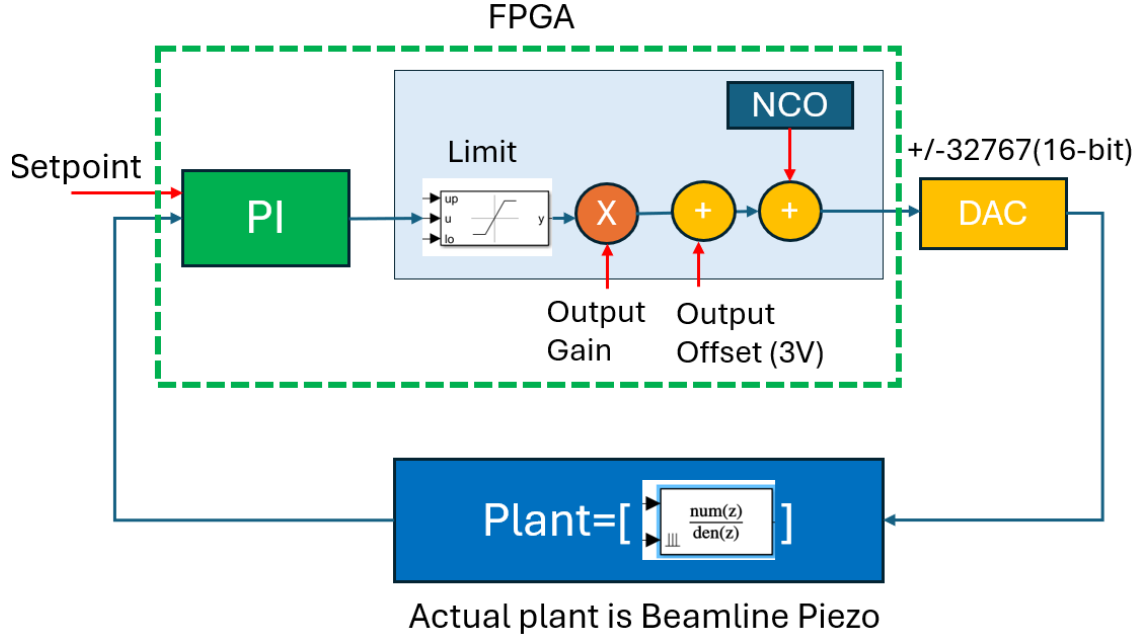


Figure 7: FPGA PI control configuration.

4. EPICS IOC and CSS display

4.1 IOC

The NSLS2EM EPICS IOC was developed based on pscDrv and quadEM [1] EPICS drivers. The pscDrv is optimized for the FPGA communications and convenient for adopting all I/O control functions. The quadEM driver was developed by APS Mark Rivers and is widely used and deployed for the beamline electrometers. The new IOC was customized with pscDrv and quadEM for the NSLS-II EM hardware. It provides several unique functions for the new beamline applications. Figure 7 illustrates the NSLS2EM EPICS IOC layout, which is implemented in hybrid mode. Most beamlines used quadEM for existing hardware, and consideration was kept for the same IOC and OPI concepts. However, the pscDrv module supports real-time fast feedback, DMA waveform data collection, and FPGA-based X-ray beam position calculation features.

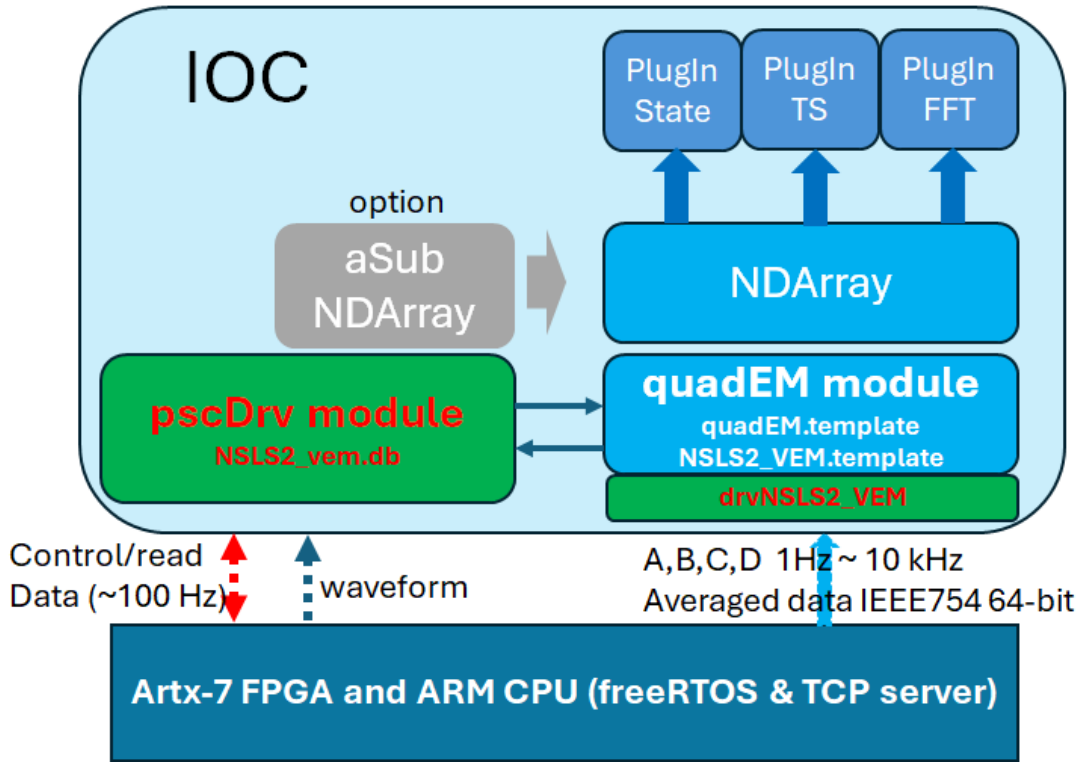


Figure 8: NSLS2EM IOC configuration.

4.2 quadEM data stream interface

Figure 9 illustrates the FA data stream structure for the quadEM IOC. The ADC clock is fixed at 378 kHz, and the FA (Fast Acquisition) rate is determined by the averaging factor applied to the ADC data.

By default, the averaging factor is set to 38, resulting in a 10 kHz output rate. However, this parameter is user-configurable, allowing adjustment down to approximately 1 kHz if lower data rates are required. In this case, at the quadEM main page, "Average per number needs to change to 10 for the right sampling frequency and monitoring Sample time 1.0E-3 for the 1 kHz sampling rate. (See quadEM top page OPI)

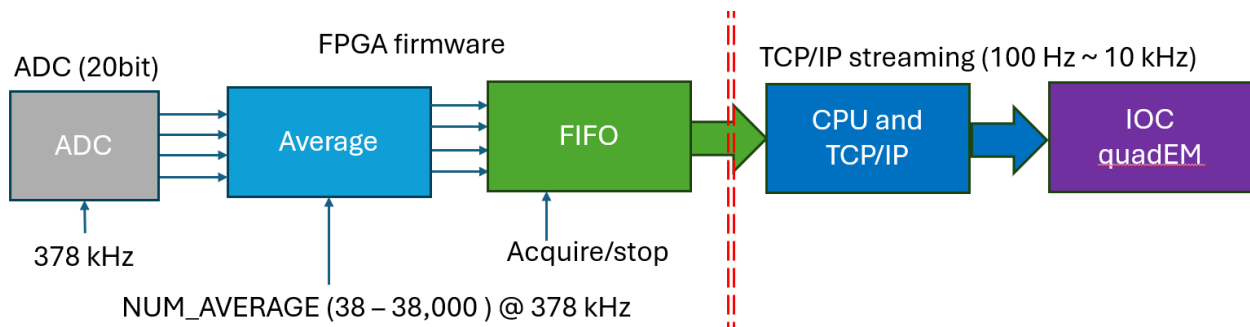


Figure 9: FA data data stream for the quadEM.

Figure 10 illustrates the data path and processing architecture for the 10,kHz data stream from the NSLS-II electrometer to the `quadEM` system.

Raw input channels labeled A, B, C, and D represent the digitized current signals from a four-quadrant photodiode or equivalent detector. Additional metadata channels—including a hardware timestamp (TS), event counter (CNT), and general-purpose I/O flags (GPIO)—are appended to each data packet.

The data stream is transmitted over a TCP/IP link and received by the `NSLS2 VEM driver`, which manages buffering, format decoding, and communication with the EPICS IOC framework. Once received, the current signals are passed to the `quadEM computePositions` module, which calculates key quantities such as:

- Individual quadrant currents (`QECurrent1--4`)
- Summed currents (`QESumX`, `QESumY`, `QESumAll`)
- Differential currents (`QEDiffX`, `QEDiffY`)
- Computed beam positions (`QEPositionX`, `QEPositionY`)

Calibration parameters, such as gain and offset for current and position channels, are applied during this computation step to ensure physical accuracy. The geometry of the detector (e.g., square or diamond layout) is also taken into account.

Finally, the processed data—including hardware timestamp, event count, and GPIO status, which are features added explicitly in the `NSLS2EM` to support flyscan operations—is made available to the `areaDetector` plugin. This plugin supports visualization and further analysis features, such as signal state monitoring, timestamp correlation, and frequency-domain analysis (FFT).

This interface architecture ensures high-throughput, low-latency delivery of real-time beam diagnostic data at a sustained rate of 10,kHz, supporting both monitoring and control applications.

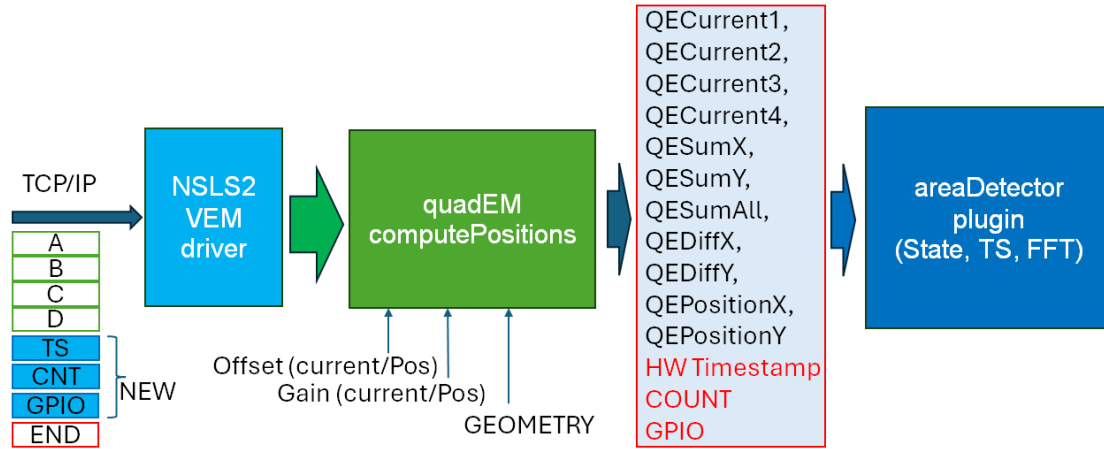


Figure 10: NSLS2EM plugin stream interface.

Figure 11 illustrates the quadEM top page. This quadEM provides plugin support and beam position calculation functionality. It maintains its own offset and gain parameters and operates independently from the FPGA-calculated position. For more information about quadEM, refer to Reference [1] for a detailed description.

Start to quadEM stream data -

- 1)Acquire and Run button engaged for stream data update
- 2)Check stream data update rate is the desired rate
- 3)Check plugin update rate and adjust average time

NSLS2_EM Quad Electrometer Rev.D

(XF:03IDB-BI(EM:1))

Model

NSLS2_EM

Firmware

NSLS2_EM RevD17072725

Sample time

5.0000E-4

Acquire

Done

Acquire

FPGA stream

Stop

Run

Acquire mode

Continuous

Continuous

Range

100 uA

0x2020202

Geometry

Diamond

Diamond

Values per reading

5

5

Averaging time

0.010

0.010

Stream update Rate

1978.5 (Hz)

1978

Plugin update Rate

99.00 (Hz)

Acquisitions(Multiple)

1

1

Acquisitions done

1448059

Read data

Read

To average

20

PID - X

Averaged

20

PID - Y

Range:A

100 uA

Range:B

100 uA

Range:C

100 uA

Range:D

100 uA

0x2020202

DAC output control

Bias

0.000

36044

DAC 0

0.000

DAC 1

0.000

DAC 2

0.000

DAC 3

0.000

NSLS2EM_bpm_mode

NSLS2EM_pscdrv_Test

Name

1

2

3

4

Current

2.798E-2

1.965E-2

1.557E-2

2.709E-2

(Sigma)

2.701E-4

2.366E-4

3.340E-4

2.226E-4

Offset

0.000E0

0.000E0

0.000E0

0.000E0

Compute

Compute

Compute

Compute

Scale (Curr./count)

1.000E0

1.000E0

1.000E0

1.000E0

Reset

Amperes

uA counts

X

Y

All

Sum

4.762E-2

4.265E-2

9.028E-2

(Sigma)

3.277E-4

4.450E-4

3.981E-4

Difference

-8.332E-3

1.152E-2

(Sigma)

3.880E-4

3.522E-4

Position

-43.7

67.5

um

(Sigma)

2.0

2.4

um

Pos. Offset

0.0

0.0

um

Pos. Offset

Compute

Compute

Pos. Scale

250.0

250.0

um

Time series

Acquire

Erase/Start

Stop

Acquiring

Plugin control

Plots

4 Currents

XY Plot

X combined

Y combined

Individual

Time domain

X combo

Y combo

Indiv.

Frequency domain

X combo

Y combo

Indiv.

Status

Ring overflows

0

Status rate

Passive

Reset

Reset

Plugins

Plugins

Asyn record

Asyn rec.

Figure 11: quadEM top page.

4.3 Gain and offset for current input

Analog input ADC gain and offset values are determined during the calibration procedure. The calibration data is stored in a text file, which includes the gain and offset values for each current input range.

When the IOC starts, it reads the calibration file and applies the corresponding gain and offset values to the process variables (PVs).

The ADC has a maximum resolution of 2^{19} bits. The offset correction is also applied to ensure accurate current measurements across all supported input ranges.

NSLS2_EM Quad Electrometer(pscdrv)

(XF:03IDB-BIEM:1)

Main	10Hz XY	10 Hz ABCD	quadEM Gain/Offset	ADC Current	FPGA-10kHz-PI	ADC-Voltage	F
------	---------	------------	--------------------	-------------	---------------	-------------	---

Gain/Offset from raw calibration data (for plugin stream data)

Range 10 mA		
	Gain	Offset
A	-515650	2702
B	-516411	1652
C	-515381	1956
D	-514540	3368

Range 100 uA		
	Gain	Offset
A	-516662	2713
B	-515388	1651
C	-515180	1961
D	-514260	3372

Range 1 mA		
	Gain	Offset
A	-515554	2702
B	-517205	1645
C	-515623	1948
D	-514223	3363

Range 10 uA		
	Gain	Offset
A	-515672	2748
B	-515990	1686
C	-514131	1992
D	-514128	3400

Range 1 uA		
	Gain	Offset
A	-519098	2742
B	-516632	1677
C	-515074	1990
D	-514224	3397

Range 100 nA		
	Gain	Offset
A	-516340	2752
B	-517870	1687
C	-518741	1998
D	-517281	3411

Figure 12: Gain and offset for quadEM stream data

4.4 Current input setting and display page

Figure 13 shows the current input control and display functions for the 4-channel input currents. Users can adjust the FA update rate and current signal input ranges. The ADC raw data sampling rate is fixed at 378 kHz (derived from the accelerator frequency), and the FA rate is the selectable averaging number based on the ADC raw rate. For example, FA rate number 38 makes FA have a 10 kHz update rate. The real FA rate frequency is measured at the RTL level and provides PV for monitoring. Actual current measurement values are displayed at the quadEM timeseries plugin page and show absolute uA values for each channel.

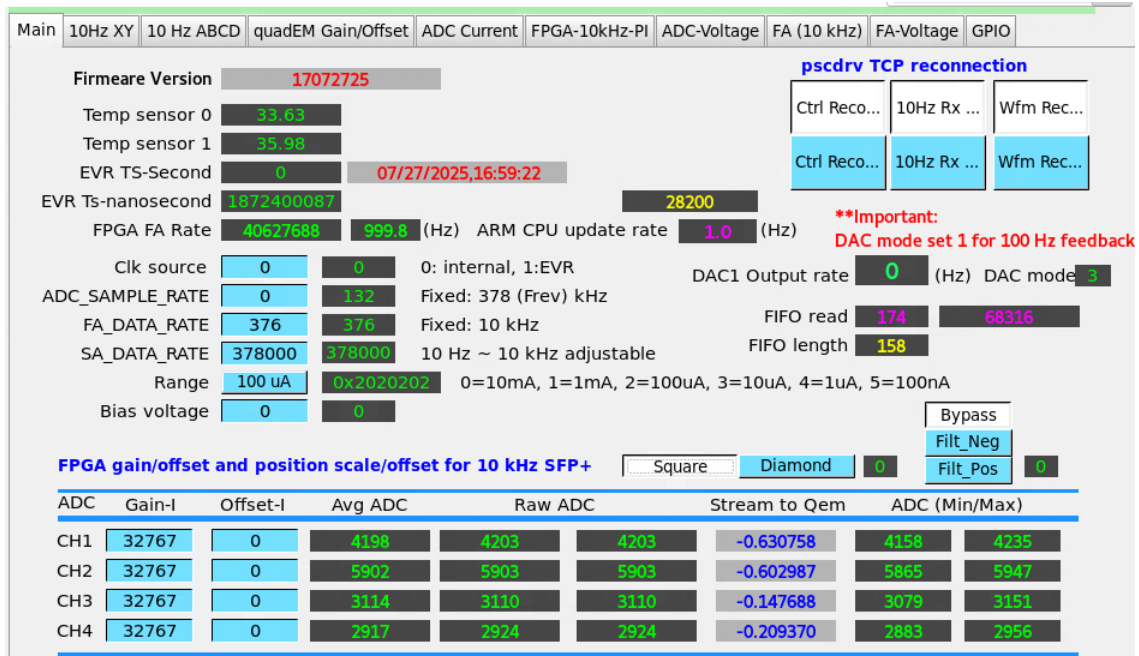


Figure 13: Current input control and monitoring page.

4.5 GPIO control and monitoring

Figure 14 shows the GPIO control OPI for the ISS beamline. The GPIO control function is special for the ISS beamline requirements. First, select the "Pulser Control ISS" mode button. Users will be able to adjust frequency and duty cycle for each TTL output channel. And press Run/Stop or set maximum clock counts. After reaching maximum counts, it will stop the clock output and show the Run/Stop status LED. After that, if you want to restart the output, press the reset button.

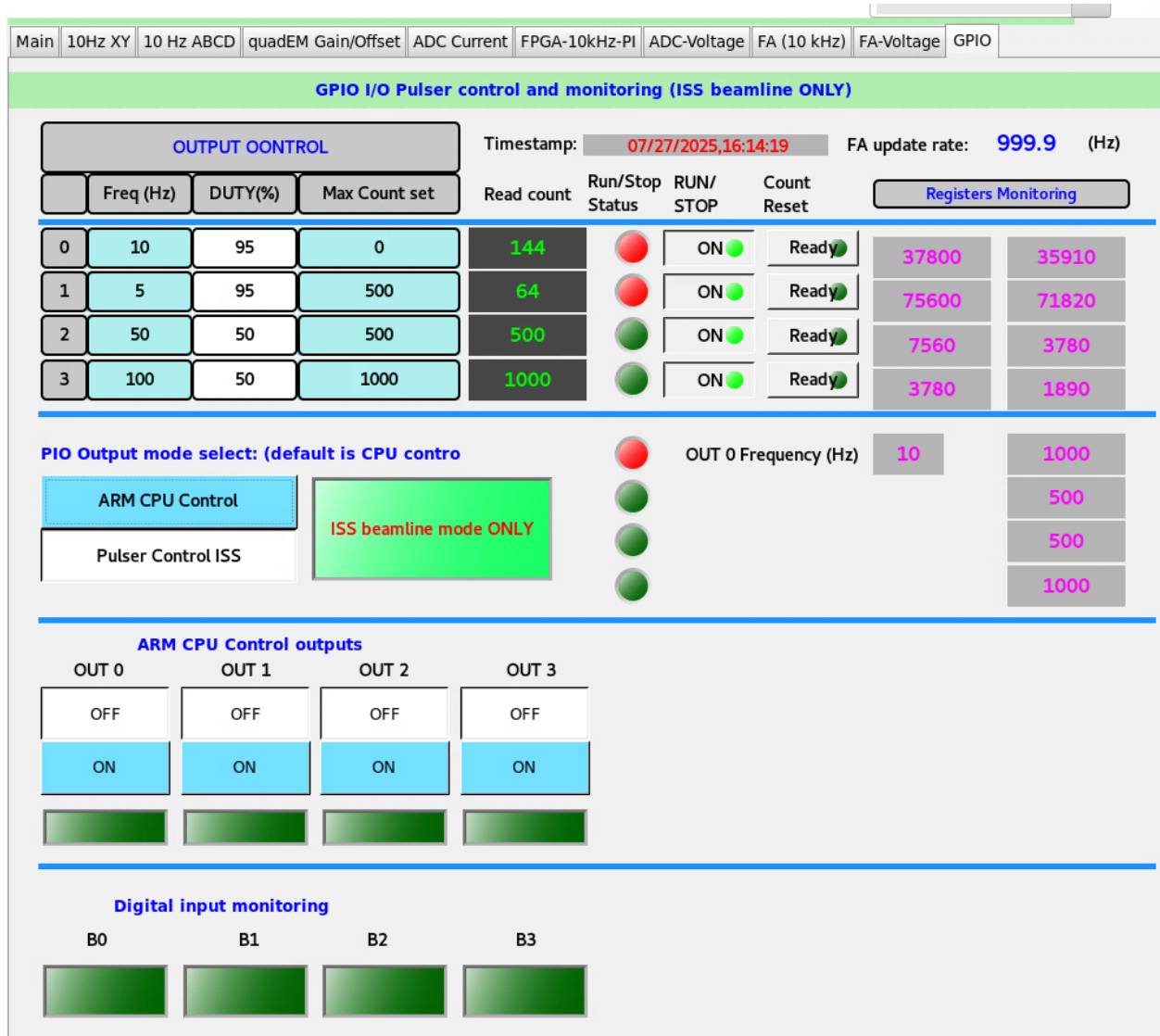


Figure 14: GPIO control page.

4.6 HDF5 file save

The HDF5 file save plugin function is provided by the quadEM standard feature, and we have added three new parameters to meet the ISS requirements.

- 1) All channel input data and calculation data are saved to a file.
- 2) EVR timestamp save to file
- 3) GPIO (4 inputs, 4 outputs) signals save to file.
- 4) Fa rate counter values are saved to file.
- 5) Used areaDetector HDF5 plugin for save files

- 6) Provides each sampling point timestamp from the accelerator timing system
- 7) Sampling rate is configurable for FA rate control. 1 kHz 10 kHz

For ISS beamline applications, they do not require beam position features or DAC output signals. The HDF5 file save function will be described in the next chapter.

Figure 15 shows the quadEM plugins for saving the HDF5 file.

XF:31ID1-BI Plugins										
Record	Output Port	Port	Addr	Enable	Blocking	Dropped	Free	Rate		
Current1	NSLS2_EM_STATS1	NSLS2_EM	0	Enable	Enable	No	0	30	100.00	More
Current2	NSLS2_EM_STATS2	NSLS2_EM	1	Enable	Enable	No	0	30	100.00	More
Current3	NSLS2_EM_STATS3	NSLS2_EM	2	Enable	Enable	No	0	30	100.00	More
Current4	NSLS2_EM_STATS4	NSLS2_EM	3	Enable	Enable	No	0	30	100.00	More
SumX	NSLS2_EM_STATS5	NSLS2_EM	4	Enable	Enable	No	0	30	100.00	More
SumY	NSLS2_EM_STATS6	NSLS2_EM	5	Enable	Enable	No	0	30	100.00	More
SumAll	NSLS2_EM_STATS7	NSLS2_EM	6	Enable	Enable	No	0	30	100.00	More
DiffX	NSLS2_EM_STATS8	NSLS2_EM	7	Enable	Enable	No	0	30	100.00	More
DiffY	NSLS2_EM_STATS9	NSLS2_EM	8	Enable	Enable	No	0	30	100.00	More
PosX	NSLS2_EM_STATS10	NSLS2_EM	9	Enable	Enable	No	0	30	100.00	More
PosY	NSLS2_EM_STATS11	NSLS2_EM	10	Enable	Enable	No	0	30	100.00	More
EVR timeST	NSLS2_EM_STATS12	NSLS2_EM	11	Disable	Disable	No	0	30	0.00	More
GPIO	NSLS2_EM_STATS13	NSLS2_EM	12	Disable	Disable	No	0	30	0.00	More
FA Count	NSLS2_EM_STATS14	NSLS2_EM	13	Disable	Disable	No	0	30	0.00	More
StdArrays	NSLS2_EM_Image1	NSLS2_EM	14	Disable	Disable	No	0	30	0.00	More
netCDF1	NSLS2_EM_FileNetCDF1	NSLS2_EM	14	Disable	Disable	No	0	30	0.00	More
Time	NSLS2_EM_TS1	NSLS2_EM	14	Enable	Enable	No	0	30	100.00	More
FileHDF5	NSLS2_EM_FileHDF1	NSLS2_EM	14	Enable	Enable	No	0	30	0.00	More

Figure 15: quadEM OPI HDF5 plugin page.

Figure 16 shows the current input data and the calculated position and differential data from the quadEM drivers. This data format was designed for X-ray BPMs; however, the ISS application needed only current data. On the other hand, each available beamline modifies the plugin database description and removes unnecessary data.

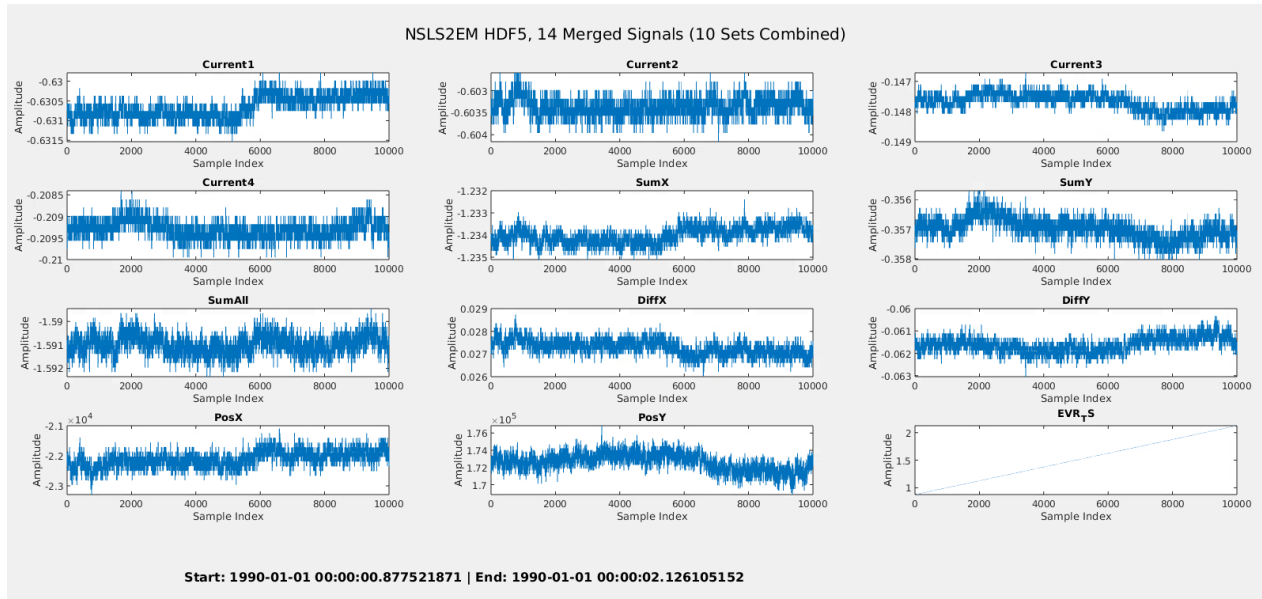


Figure 16: Current readout from HDF5 file. (For the test purpose, used Matlab to read HDF5 and plot)

Figure 17 shows GPIO data from the HDF5 file; bits 0 to 3 are input signals, and bits 4 to 8 are output clock signals. Each sampling point is correlated with an EVR timestamp, which utilizes a 64-bit data format. An important consideration is the FA sampling rate, GPIO clock output frequency, and duty configuration. For example, the ISS beamline FA rate is configured with a 1 kHz update rate, which means the quadEM sampling rate is also 1 kHz, and the EVR timestamp updates at 1 kHz (1 ms) for the HDF5 file. In this case, the GPIO clock settings above are set to 100 Hz and a 95 percent duty cycle, which causes a missing point because 1 kHz FA sampling is insufficient for maintaining a 5 percent low status on GPIO. So with the 1 kHz update rate configuration, the duty cycle is less than 90 percent for the 100 Hz GPIO clock output.

Example:

Let's say FA update is 1 kHz (1 ms) and GPIO-0 is set to 100 Hz and 95 % duty: register value for frequency 3780, and duty 3591

$F_s = 1/378000 = 2.645502645502646 \times 10^{-6}$ sec, 2.6455 us, for GPIO control of one clock.

Clock register $3780 * 2.6455 = 10,000$ us = 10 ms (100 Hz)

Duty register (high 95 %): $3591 * 2.6455 = 9.495$ ms

Duty low 5 % = $189 * 2.6455$ us = 0.5 ms

So, it will not detect low status from the quadEM 1 kHz FA sampling rate. In this case, we need to increase the FA rate to 10 kHz.

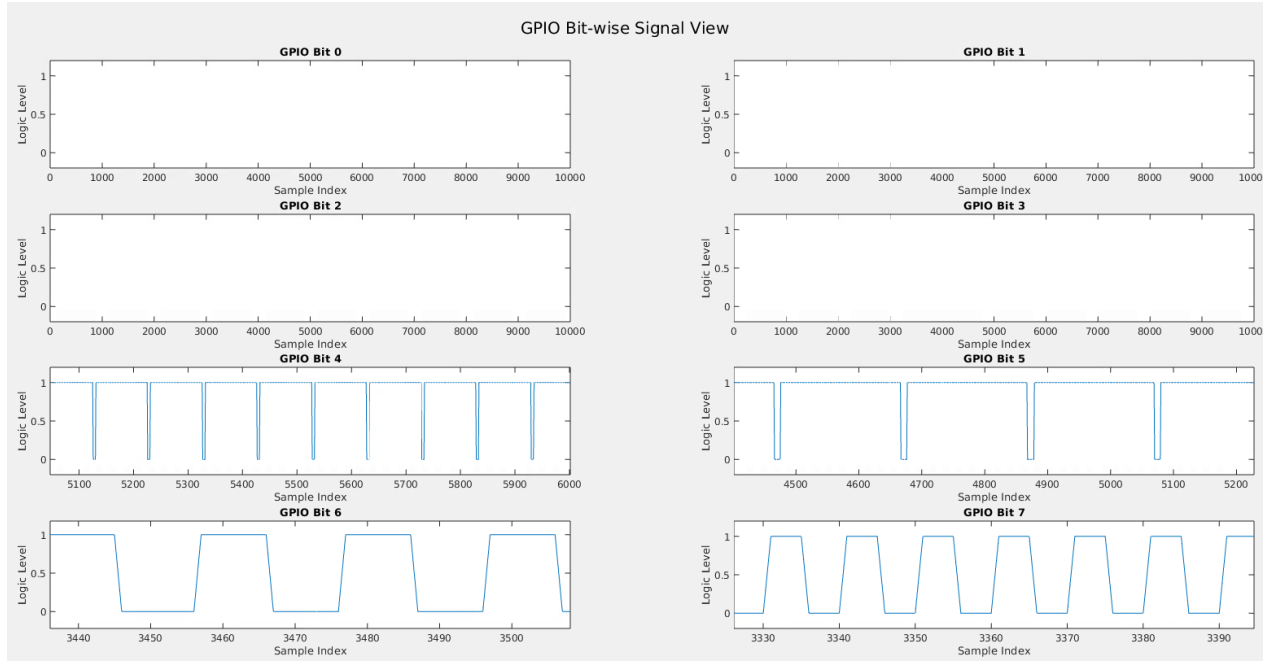


Figure 17: GPIO data readout from HDF5 file. (Bit 0-3 are inputs and currently no input signals)

4.7 DAC Output control

It is important for users to understand how the DAC control mode operates.

.....

4.8 PI control

The PI is a special control mode and requires coordination with the beamline user for testing.

5. Additional NSLS2EM functions

The new version of the NSLS2EM firmware provides additional functions for the beamline application.

- 1) 378 kHz 300 k points waveform PVs for 4-channel ADC.
- 2) 10 kHz 50 k points waveform PVs for 4-channel FA data.
- 3) 4-channel voltage inputs.
- 4) NCO sine/cos waveform generator for DAC output.

- 5) Real-time stream data display and FFT display by quadEM.

6. Analog input calibration

The analog input calibration procedure will be provided in a separate document. After assembling new hardware units, all channels will be calibrated, and a coefficient text file will be generated accordingly.

7. Summary

The FPGA firmware version 17 supported ISS requirement functions and was available for installation at the beamline. After installation and beta version testing, we made available remote firmware update functions for any new requirements. The quadEM OPI functions will need to ask to contact each beamline assistant engineer (ISS is assigned to Jakub)

8. Appendix

8.1 The ISS (8-ID) Beamline requirement

The ISS (8-ID) beamline requested new NSLS2EM hardware for the hutch-B installation. On July 22, during the NSLS2EM technical meeting, Jorge and Akhil introduced the 8-ID EM requirements and schedule. See the appendix for the provided layout and requirements. The DSSI working group is actively supporting the new NSLS2-EM firmware, EPICS IOCs, and OPI development, and is considering an Ansible role to upgrade FPGA firmware remotely. For ISS beamline requirements were reviewed and implemented by new firmware. [Figure 18](#) shows requirement of the ISS beamline.

ISS EM requirements

- >10 kHz acquisition rate with timestamping
- 4 channels for current readout (3 ICs, 1 PIPS)
- Signal range nA to 100s uA (10^{-4} to 10^{-9} A)
- TTL pulse output >1 kHz with variable duty cycle
- Voltage input for TTL signals with timestamping
- Software: EPICS IOC, CSS screen
- Start and stop acquisition with EPICS PVs
- Timeline – start of the September 2025-3 cycle

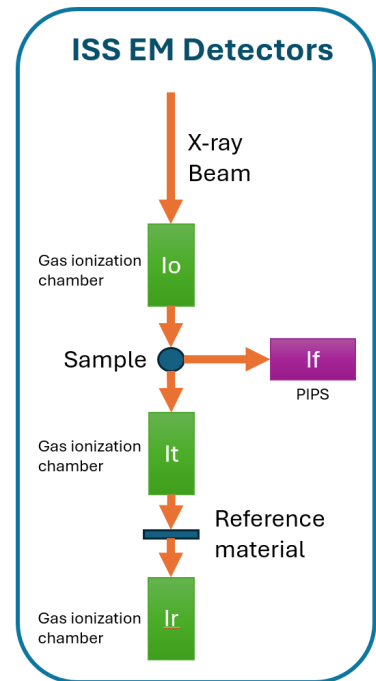


Figure 18: ISS beamline requirements. (from Jorge July,22/2025)

Installed 8/18/25

IOC xf08idb-ioc2.nsls2.bnl.local

WS xf08id-sw1.nsls2.bnl.local

EM 10.66.58.43

Test IOC: /epics/iocs/nsls2em-kha/iocBoot

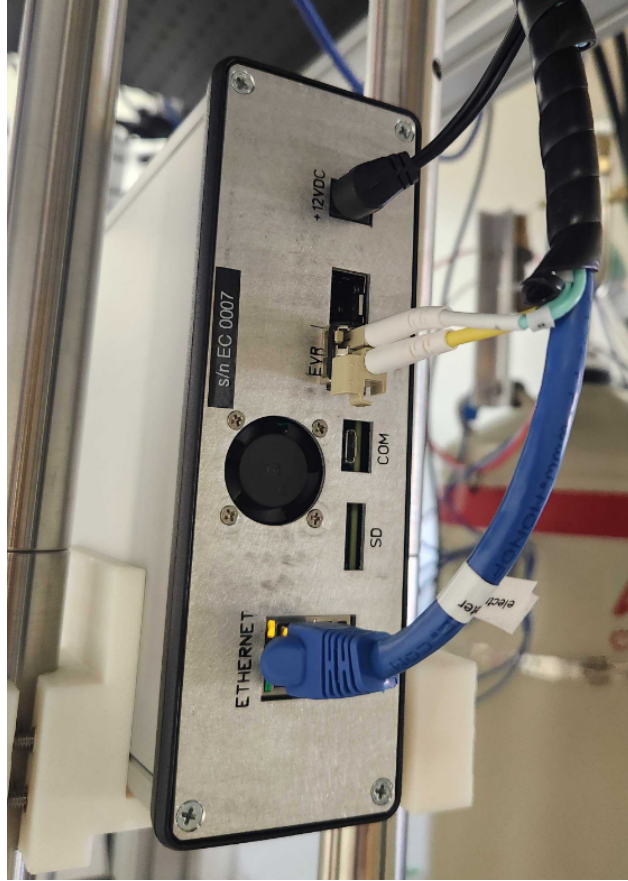


Figure 19: ISS hutch-b installation

8.2 The HXN (3-ID) Beamline requirement and status

Currently, HXN (3-ID) Hutch-C is operating with the new firmware and IOC, and the BPM in Hutch-C is configured for monitoring mode only. In contrast, the BPM in Hutch-B requires local feedback functionality for both X and Y axes. The PI-based local feedback was successfully demonstrated at Lab3 using both open-loop and closed-loop tests. DSSI and the HXN team plan to test the PI closed-loop feedback with the X-ray beam and evaluate its overall performance.

The HXN (3-ID) Hutch-C uses an ISO model electrometer (EM) based on older hardware, and currently, no working spare unit is available. The associated IOC is not compatible with the new EM hardware. Figure 20 shows the NSLS2EM system installed in Hutch-B at beamline 3-ID (HXN).

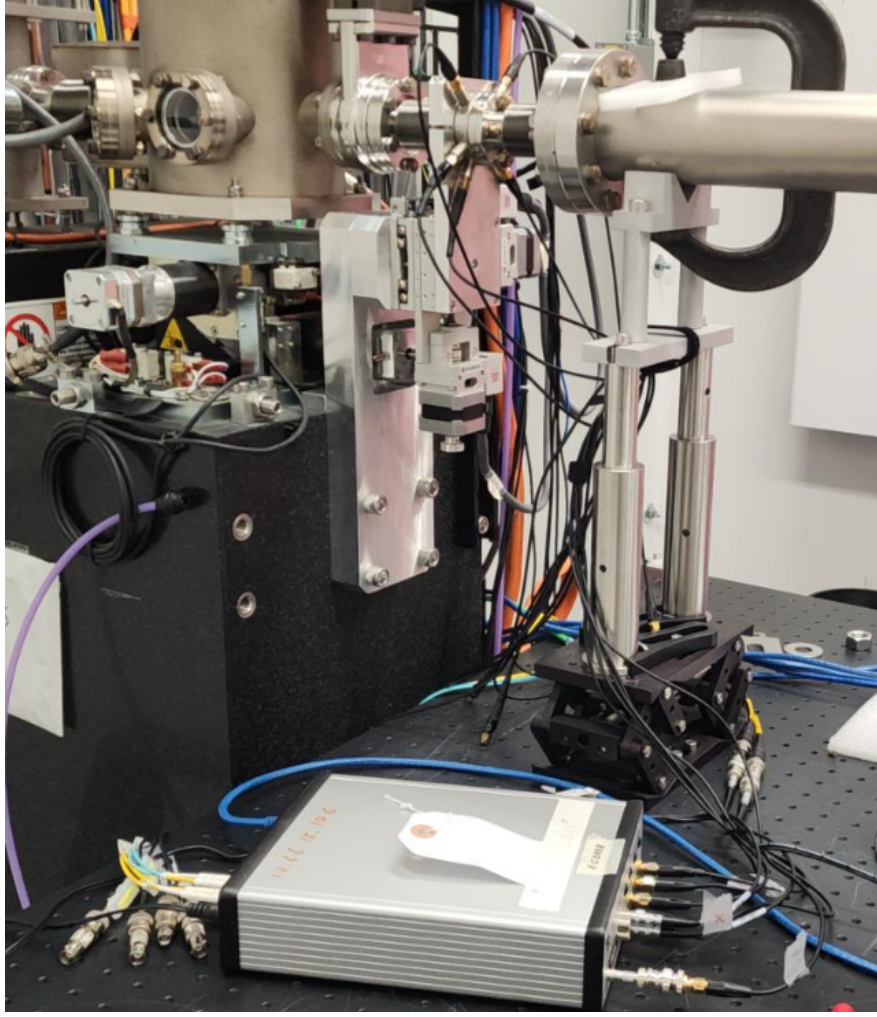


Figure 20: HXN hutch-b installation (x/y DAC offset=3V, bias=1V)

Table 4: HXN(3-ID) Host and IOC hostname

System	Hostname
Workstation	xf03idc-ws1.nsls2.bnl.local
IOC Host	xf03idc-ioc2.nsls2.bnl.local

Device: Hutch-B 10.66.18.146 Hutch-B 10.66.18.147

Table 5: NSLS2EM IOC and Module Paths

IOC directory	Location	Type	IOC Path
/epics/modules/NSLS2EM-V17	Hutch-C	ISO Old EM	/epics/iocs/nsls2em-b-v17
/epics/modules/NSLS2EM-V18	Hutch-B	New EM	/epics/iocs/nsls2em-b-v18

On August 1, 2025, a beam study was conducted to evaluate the FPGA-based PI control functionality. The test confirmed that both X and Y plane feedback loops operated successfully.

Summary: 3-ID Local Feedback Study with New Electrometer

- The open-loop transfer function of the system was measured on both the X and Y planes.
- The loop was then closed, and PID parameters were adjusted to evaluate the system's ability to suppress noise.
- A sinusoidal excitation was applied to the piezo actuator. The feedback system effectively suppressed the induced vibration up to approximately 30 Hz on the horizontal (X) plane and 20 Hz on the vertical (Y) plane.
- The PID parameters used during the test were:
 - $K_{p_x} = 3.2$, $K_{i_x} = 0.08$
 - $K_{p_y} = -4$, $K_{i_y} = -0.4$
- These parameters may require further tuning, as the current settings may amplify high-frequency vibrations.
- Overall, the feedback system operated as expected, and the new electrometer (EM) demonstrated effective low-frequency vibration suppression.

8.3 C09 (CID) front-end XBPM Requirement

The storage ring (SR) C09 used an XDS OXFORD blade XBPM and recommended the I400 electrometer [4]. We will instead standardize on the NSLS2EM electrometer due to its stronger EPICS integration, built-in X-ray beam-position calculation support, and related operational benefits.

Device IP : 10.0.142.50

8.4 The Biology Beamlines Requirement

8.5 The 29-ID ARI/SXN Beamlines Requirement

8.6 Other Beamlines Requirement

9. Installation

9.1 Power

Power used DC +12V and DSSI will provide the correct AC/DC adapter with NSLS2EM hardware.

9.2 UART serial console

To configure the IP address via serial, connect to the Zynq UART serial port using a serial terminal set to 115200 bps. Then, type "h" to display the list of supported commands. For Windows OS, I used TeraTem terminal. Windows 11, maybe need to install the CP210x VCP Windows driver at [5].

9.3 Ethernet IP setting

The NSLS2EM IP setting is available by serial console command or by copying the "ethernet.cfg" file from the PC to the SD card. Initially, we provide complete IP settings for each beamline that is pre-assigned for the NSLS2EM hardware. At serial console type 'h' command and show available commands. example: "ipget" for IP print, "ipset" for the new IP setting.

9.4 FPGA firmware update

To remotely update the firmware, the TFTP protocol is used. The TFTP server on the NSLS2EM ARM CPU is always in standby mode, allowing users to update the 'BOOT.BIN' file—which includes both the FPGA firmware and ARM application whenever needed. Figure 21 illustrates the process of updating the BOOT.BIN file using TFTP client commands. The new BOOT.BIN will overwrite the existing file on the SD card.


```
tftp> [kha@xf31id1-lab3-pandadev1 ns1s2em_dma]$ tftp 10.69.58.67
tftp> bin
tftp> put BOOT.bin
tftp> █
```

Figure 21: Firmware update using tftp client. (binary mode, BOOT.bin size is 64 Mbyte)

After updating the BOOT.BIN file, a system reboot is required to load the new firmware. The IOC provides a reboot process variable (PV), allowing users to initiate the reboot through an EPICS PV interface.

The system can be rebooted via the following console command:

```
$caput XF:03IDB-BI{EM:1}SysReboot-Sp 1234567
```

9.5 EPICS modules

Below epics modules for beamline configurations from Lab3 "xf31id1-lab3-ioc1"

```
du -h --max-depth=1 /epics | sort -h
```

```
4.0K  /epics/common
1.7G  /epics/iocs
3.1G  /epics/modules
6.1G  /epics/bundles
36G   /epics/utils    → /epics/utils/rhel8-epics-config source for build
47G   /epics
```

```
du -h --max-depth=1 /usr/lib/epics | sort -h
```

```
28K   /usr/lib/epics/startup
56K   /usr/lib/epics/cfg
80K   /usr/lib/epics/protocol
108K  /usr/lib/epics/iocBoot
172K  /usr/lib/epics/pmc
512K  /usr/lib/epics/templates
1012K /usr/lib/epics/req
1.4M  /usr/lib/epics/configure
5.2M  /usr/lib/epics/db
6.4M  /usr/lib/epics/include
48M   /usr/lib/epics/dbd
390M  /usr/lib/epics/lib
663M  /usr/lib/epics/bin
1.1G  /usr/lib/epics
```

10. Reference

- [1] <https://epics-modules.github.io/quadEM/quadEMDoc.html>
 - [2] <https://teratermproject.github.io/index-en.html>
 - [3] <https://www.avnet.com/americas/products/avnet-boards/avnet-board-families/picozed/>
 - [4] <https://xds-oxford.com/products/detectors-diagnostics/beam-position-monitors/blade-bpm/>
 - [5] <https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers?tab=downloads>
- CP210x VCP Windows - works Windows 10 and 11

11. FPGA firmware revision control

11.1 Vivado

Vivado 2022.2 Firmware version: 17:07/31/25

- ISS requirements added: GPIO, HW timestamp, HDF5 file
- HXN requirements added: BPM x/y position calculation, PI controller, Bias control, DAC output control

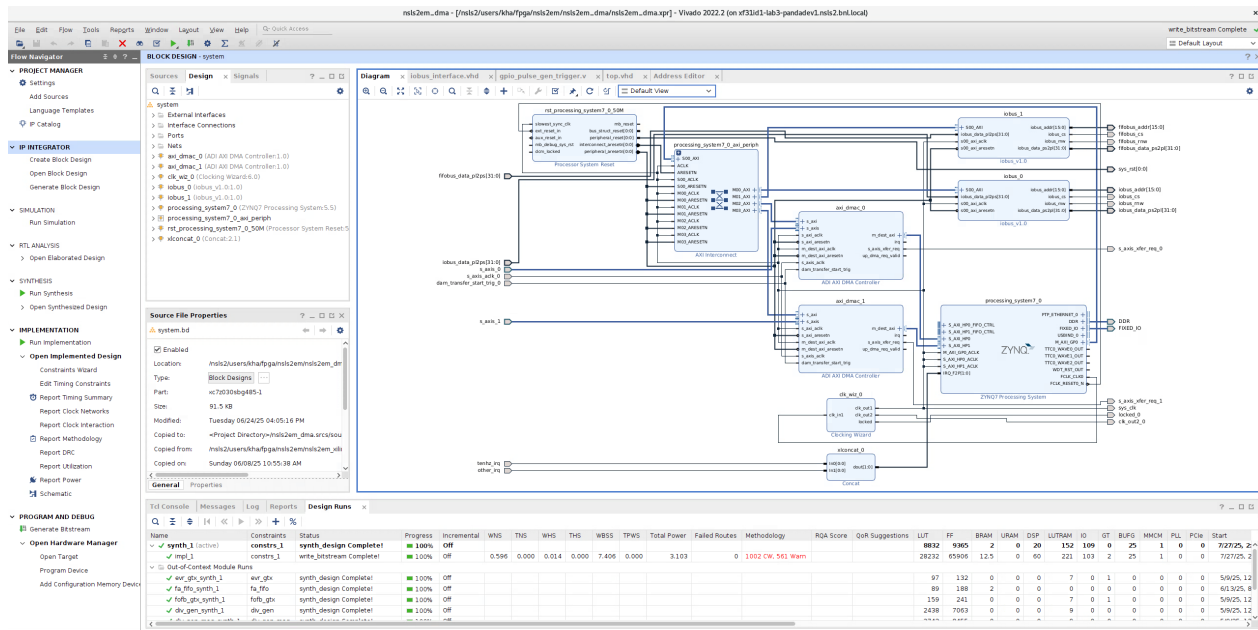


Figure 22: Vivado 2022.2 project top

11.2 freeRTOS lwIP BSP configuration

LIBRARY_NAME	lwip211
LIBRARY_VER	1.8
PROC_INSTANCE	ps7_cortexa9_0
api_mode	SOCKET_API
dhcp_does_arp_check	true
lwip_dhcp	true
mem_size	524288
memp_n_pbuf	1024
memp_n_tcp_seg	1024
memp_num_netbuf	4096
n_rx_descriptors	512
n_tx_descriptors	512
pbuf_pool_size	16384
tcp_ip_rx_checksum_offload	true
tcp_ip_tx_checksum_offload	true
tcp_rx_checksum_offload	true
tcp_snd_buf	65535
tcp_tx_checksum_offload	true
tcp_wnd	65535

12. Known Issues

12.1 Hardware

The C107 capacitor, 220 pF, caused 2 MHz oscillation issues and was replaced with a 2 pF.

12.5 OPI

FFT plugin update

The FFTAbsValue output from the quadEM FFT plugin is not updating. This issue does not affect all channels—only specific ones are impacted (8/1/25 during study). Not sure IOC or OPI.

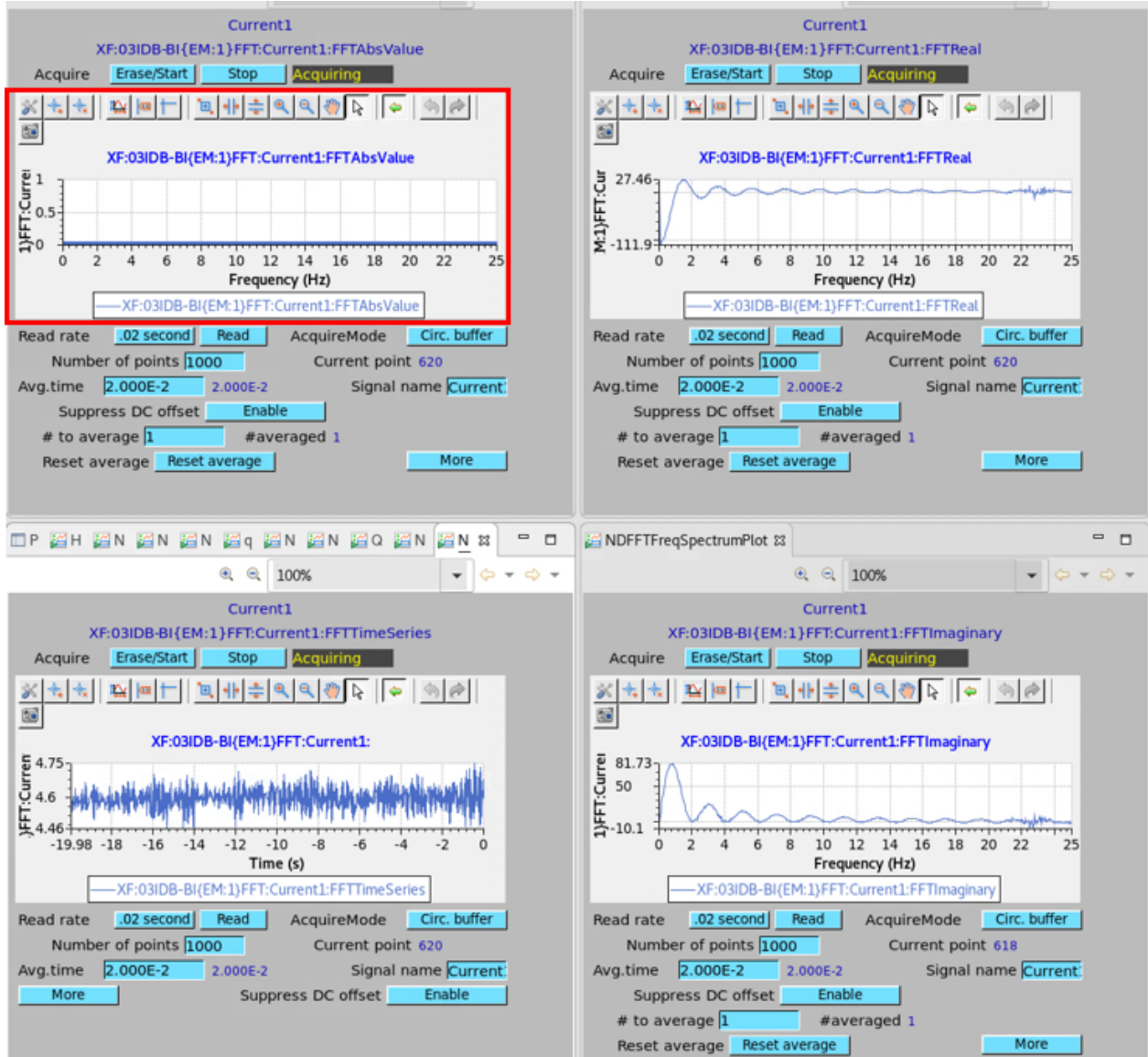


Figure 24: FFT plugin and X position FFTAbs issues (3-ID Hutch-B)

FFT peak value

During operation, the HXN hutch-C Y position FFT shows an unexpected 360 Hz pick value, and plotting the ADC raw data in Matlab did not show a 360 Hz peak.

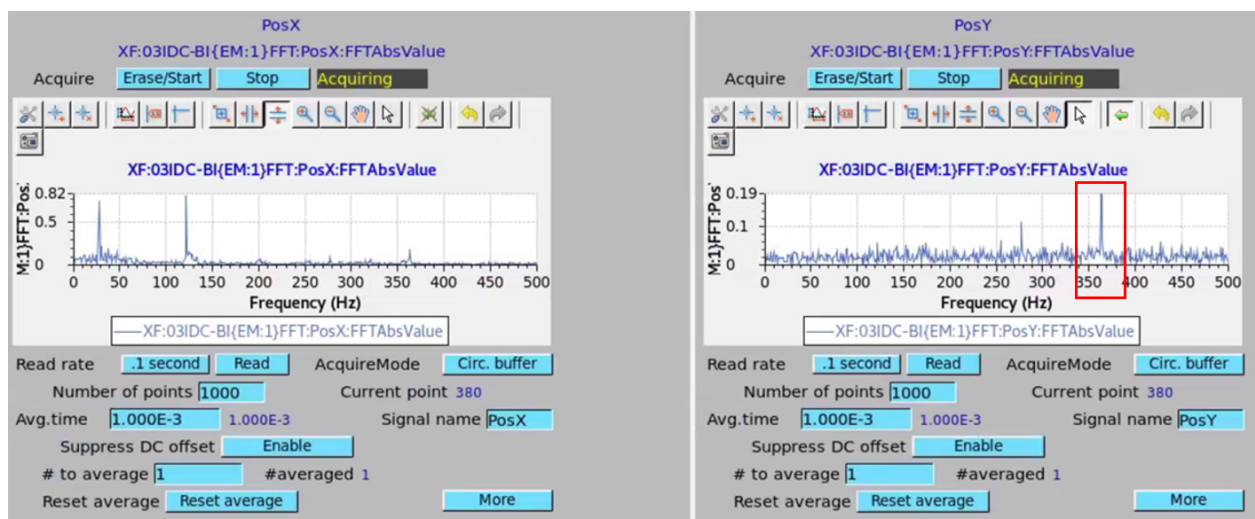


Figure 25: FFT unknown peak at HXN hutch-C, (Y position 360 Hz peak)

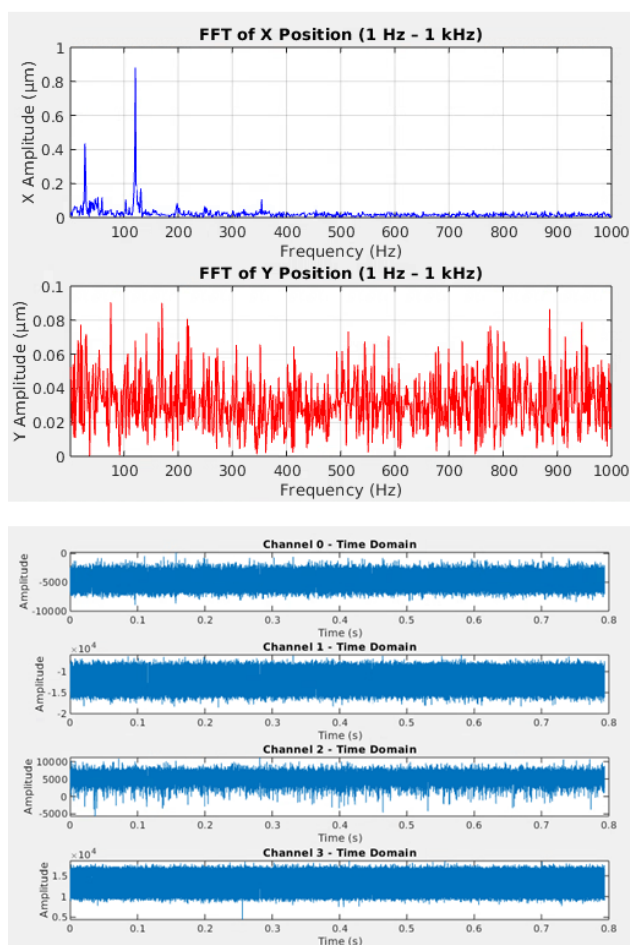


Figure 26: FFT offline calculation, (No Y position 360 Hz peak)